

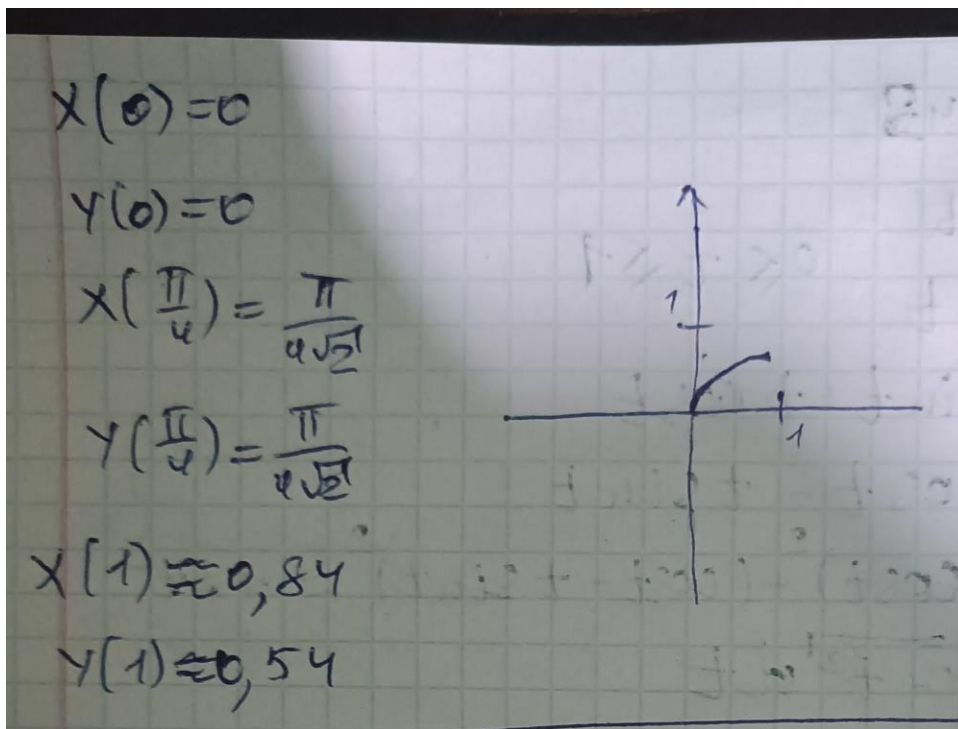
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$x = 3 \sin t$
 $y = 3 \cos t$
 $0 < t < 2\pi$
 $f'(x) = -\frac{3}{2} \tan t$
 $f''(x) = -\frac{3}{4} \sec^3 t$
 $-\frac{3}{4} \sec^3 t = 0$
 $t = \pm \frac{\pi}{2} + \pi n$
 $t \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

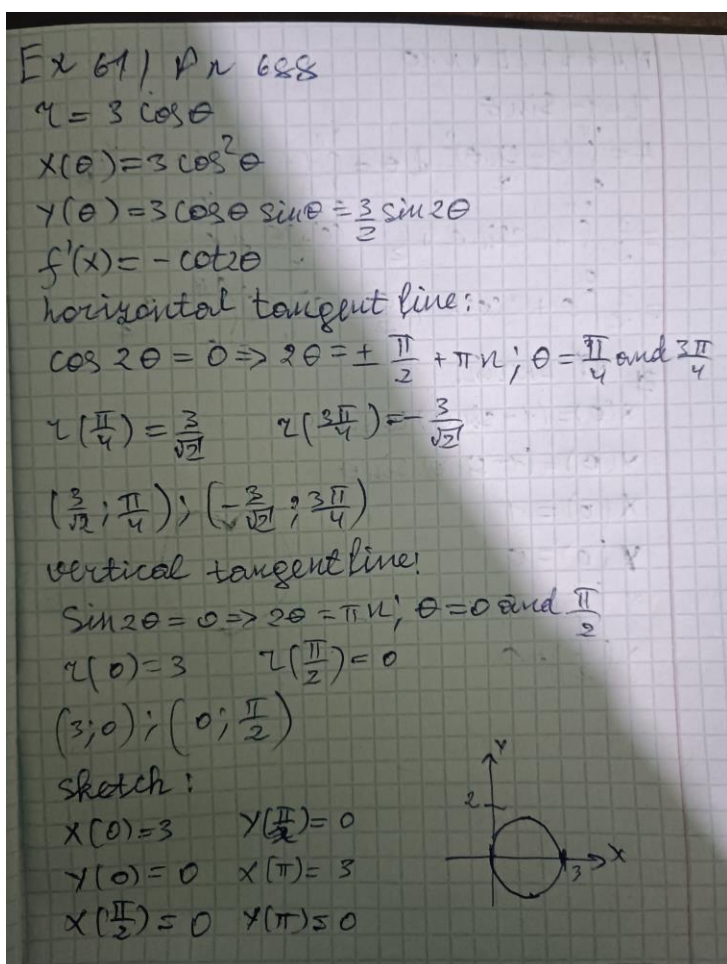
$x(0) = 0$
 $x(\pi) = 0$
 $x(\frac{\pi}{2}) = 3$
 $x(\frac{3\pi}{2}) = -3$
 $y(\pi) = -3$
 $y(\frac{\pi}{2}) = 3$
 $x(\frac{\pi}{4}) = \sqrt{2}$
 $y(\frac{\pi}{4}) = \frac{3}{\sqrt{2}}$

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$x = t \sin t$
 $y = t \cos t$
 $0 \leq t \leq 1$
 $x'(t) = \sin t + t \cos t$
 $y'(t) = \cos t - t \sin t$
 $(\sin t + t \cos t)^2 + (\cos t - t \sin t)^2 = 1 + t^2$
 $\text{length} = \int_0^1 \sqrt{1+t^2} dt$
 $\int_0^1 \sqrt{1+t^2} dt = \int_0^1 \sqrt{1+\tan^2 x} \sec^2 x dx = \int_0^1 \sec^3 x dx =$
 $= \int \sec x \cdot \sec^2 x dx = \tan x \cdot \sec x - \int \sec x \tan^2 x dx =$
 $= \tan x \cdot \sec x - \int \sec^3 x dx + \int \sec x dx$
 $\int \sec^3 x dx = \tan x \cdot \sec x - \int \sec^3 x dx + \int \sec x dx$
 $\int \sec^3 x dx = \frac{1}{2} \tan x \sec x + \frac{1}{2} \ln(\sec x + \tan x)$
 $\sec x = \sqrt{1+t^2}$
 $\int \sqrt{1+t^2} dt = \frac{1}{2} t \sqrt{1+t^2} + \frac{1}{2} \ln(\sqrt{1+t^2} + t)$
 $\int_0^1 \sqrt{1+t^2} dt = \frac{1}{2} \sqrt{2} + \frac{1}{2} \ln(\sqrt{2} + 1) \approx 1.1478$



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$$r = 1 - \sin \theta$$

$$A = \int_0^{2\pi} \frac{(1 - \sin \theta)^2}{2} d\theta = \int_0^{2\pi} \frac{1 - 2\sin \theta + \sin^2 \theta}{2} d\theta =$$

$$= \frac{1}{2} \int_0^{2\pi} 1 - 2\sin \theta + \frac{1}{2} - \frac{\cos 2\theta}{2} d\theta =$$

$$= \frac{1}{2} \left[\theta + 2\cos \theta + \frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{2\pi} = \frac{3\pi}{2}$$

Sketch:

$$x(\theta) = (1 - \sin \theta) \cos \theta$$

$$y(\theta) = (1 - \sin \theta) \sin \theta$$

$$x(0) = 1$$

$$y(0) = 0$$

$$x\left(\frac{\pi}{2}\right) = 0$$

$$y\left(\frac{\pi}{2}\right) = 0$$

